

Intracranial Activity of Trastuzumab Deruxtecan in HER2-Low Breast Cancer with Brain Metastases



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INTRODUCTION

Human epidermal growth factor receptor 2 (HER2)-low breast cancer, defined as HER2 immunohistochemistry (IHC) 1+ or IHC 2+/in situ hybridization-negative, represents a newly targetable subtype comprising nearly half of all breast cancers. Brain metastases are a common and morbid complication of metastatic breast cancer, yet patients with active central nervous system (CNS) disease are frequently excluded from clinical trials. Trastuzumab deruxtecan (T-DXd), an antibody-drug conjugate with a cleavable linker and potent topoisomerase I inhibitor payload, has demonstrated substantial intracranial activity in HER2-positive disease. However, the intracranial efficacy of T-DXd in patients with HER2-low breast cancer remains incompletely characterized.

METHODS

A systematic literature search was conducted across PubMed/MEDLINE, Embase, CENTRAL, ClinicalTrials.gov, and Web of Science from database inception through January 1, 2025. A total of 2,275 records were identified and screened. Eligible studies included randomized controlled trials and prospective or retrospective clinical studies evaluating trastuzumab deruxtecan in patients with HER2-low metastatic breast cancer and measurable brain metastases. Studies were required to report intracranial objective response rate (IC-ORR), defined as complete or partial intracranial response among CNS-evaluable patients, assessed by study-defined criteria (RECIST 1.1 or RANO-BM).

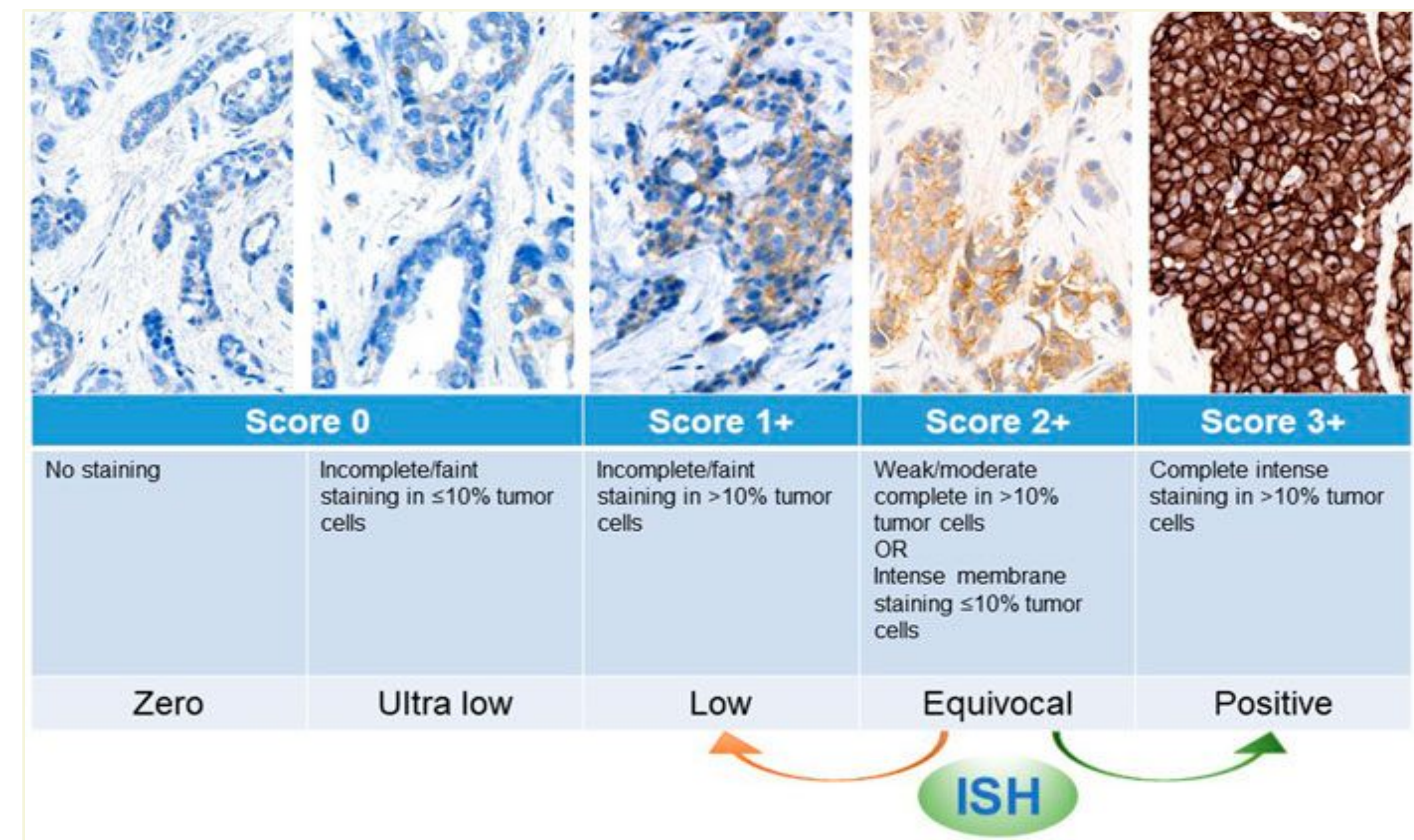


Figure 1 HER2 expression scoring by immunohistochemistry (IHC)

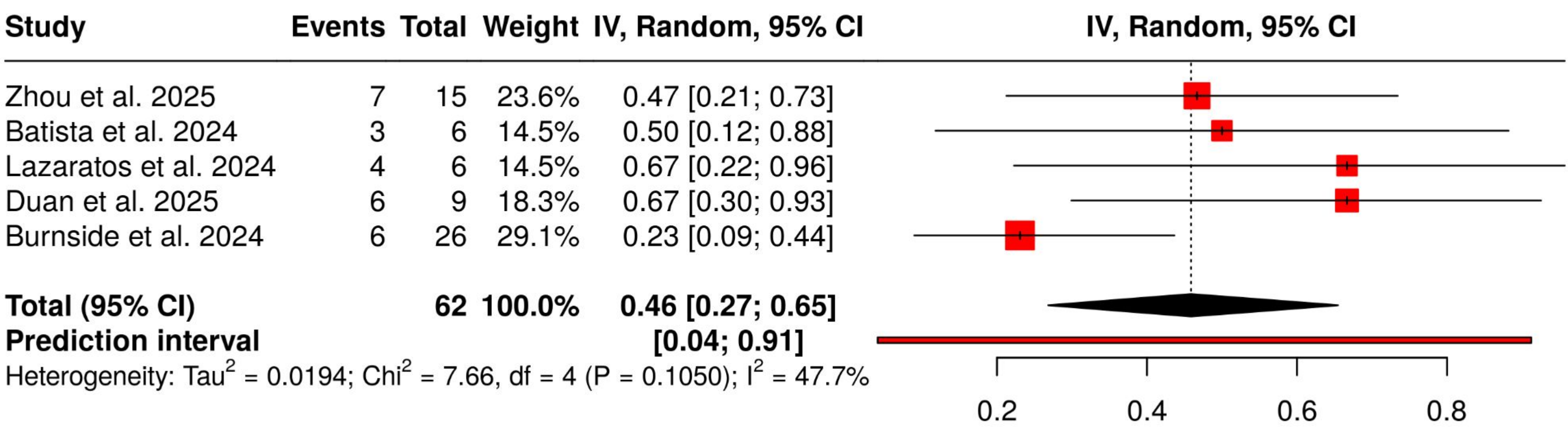


Figure 2 Forest Plot of Intracranial Objective Response Rate (IC-ORR)

Study	Year	CNS-Evaluable Patients (N)	Intracranial Responders (CR+PR)	IC-ORR % (95% CI)
Zhou et al.	2025	15	7	46.7 (21.0–73.0)
Batista et al.	2024	6	3	50.0 (12.0–88.0)
Lazaratos et al.	2024	6	4	66.7 (22.0–96.0)
Duan et al.	2025	9	6	66.7 (30.0–93.0)
Burnside et al.	2024	26	6	23.1 (9.0–44.0)
Pooled estimate (random effects)	—	62	26	46.0 (27.0–65.0)*

Table 1 Summary of Intracranial Response Outcomes

RESULTS

Five studies comprising 62 patients with HER2-low metastatic breast cancer and brain metastases were included. Using a random-effects proportional meta-analysis with Freeman-Tukey double arcsine transformation, the pooled intracranial objective response rate (IC-ORR) was 46% (95% CI, 27–65%). Between-study heterogeneity was moderate ($I^2 = 47.7\%$, $p = 0.11$), with consistent directionality of intracranial activity observed across included cohorts.

CONCLUSIONS

Across five studies, T-DXd demonstrated clinically meaningful intracranial activity in HER2-low metastatic breast cancer with brain metastases, yielding a pooled IC-ORR of 46%. This provides one of the first quantitative benchmarks of CNS response in HER2-low disease, where evidence has been limited to small, heterogeneous cohorts. These findings support dedicated prospective trials in HER2-low patients with active CNS disease and standardized intracranial endpoints to refine patient selection and expected benefit.

CLINICAL IMPORTANCE:

Brain metastases are common and morbid in metastatic breast cancer, yet patients with active CNS disease are frequently excluded from clinical trials, leaving clinicians with limited evidence to guide therapy.

HER2-low breast cancer represents nearly half of all breast cancers, but until recently was not considered a targetable subtype.

This meta-analysis provides one of the first pooled estimates of intracranial response in HER2-low disease, demonstrating a clinically meaningful IC-ORR of ~46% with trastuzumab deruxtecan.

These findings suggest trastuzumab deruxtecan has true CNS activity beyond classic HER2-positive disease, supporting its consideration in HER2-low patients with brain metastases.

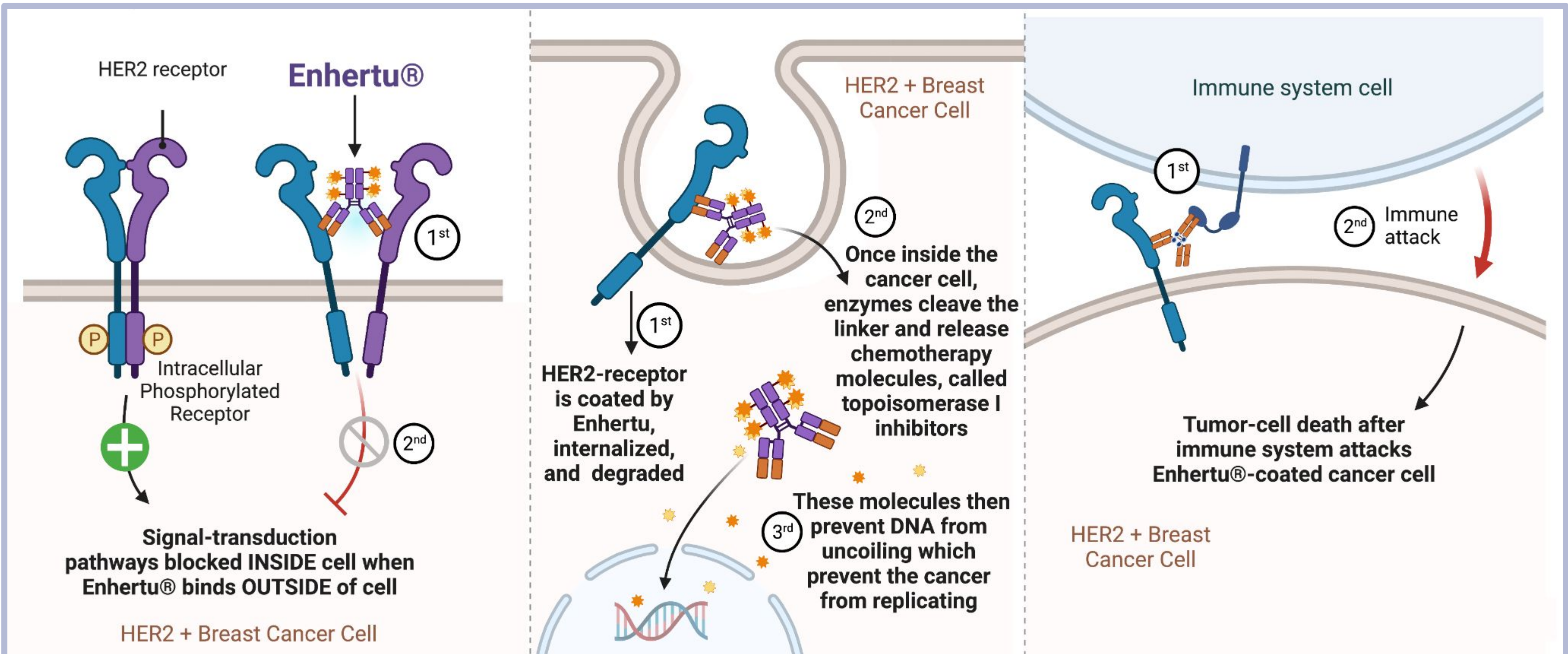


Figure 3 Mechanism of action of trastuzumab deruxtecan (T-DXd)